STACK OVERFLOW DEVELOPER SURVEY ANALYTICS: TECHNOLOGY TRENDS, DASHBOARD INSIGHTS & DEMOGRAPHIC ANALYSIS

TRANSFORMING RAW SURVEY DATA INTO ACTIONABLE INSIGHTS THROUGH DATA WRANGLING, DASHBOARDING, AND ANALYTICAL STORYTELLING

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TOOLS & TECHNOLOGIES:

EXCEL | POWER QUERY | SQL | PYTHON | JUPYTER | COGNOS ANALYTICS

SUMMARY

- Built a multi-dashboard analytics solution from survey data by cleaning multi-select technology fields, normalizing the dataset, modelling relationships in Cognos Analytics, and producing insights on developer tool adoption, future preferences, and demographics

BUSINESS / ANALYTICAL QUESTION

- What technologies are developers using today, what do they want to use next year, and how do those patterns vary within the broader respondent demographic profile?

DATASET

- Main dataset: Stack Overflow Developer Survey responses
- Supporting artefacts included:
  - job-postings.xlsx outputs
  - popular-languages.csv outputs
  - dashboard screenshots and analytics results compiled into Data Analyst Capstone Project Report

TOOLS USED

- Excel
- Power Query
- Python
- Jupyter Notebooks
- SQL/Relational Databases
- Cognos Analytics

PROCESS (WHAT I DID)

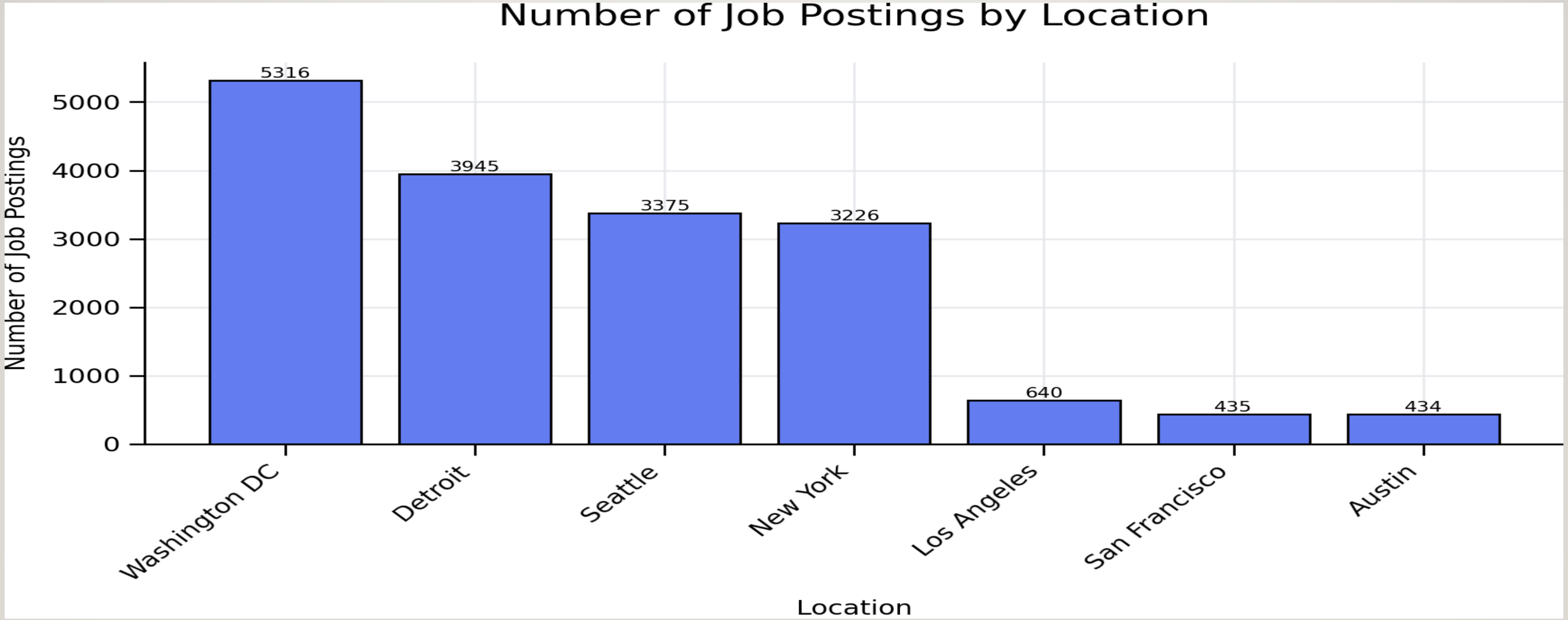
- **Data Preparation**
- Identified survey fields containing semicolon-separated multi-select technology values
- Split the values into columns in Excel
- Unpivoted the split columns into rows using Power Query
- Created **8 normalized CSV files** for separate technology dimensions
- Added the original survey table back for demographics
- **Data Modelling**
- Built a Cognos data module
- Linked normalized technology tables and the original survey table using ResponseId
- Structured the model to support dashboard analysis of languages, databases, platforms, web frameworks, and demographics.

PROCESS (WHAT I DID)

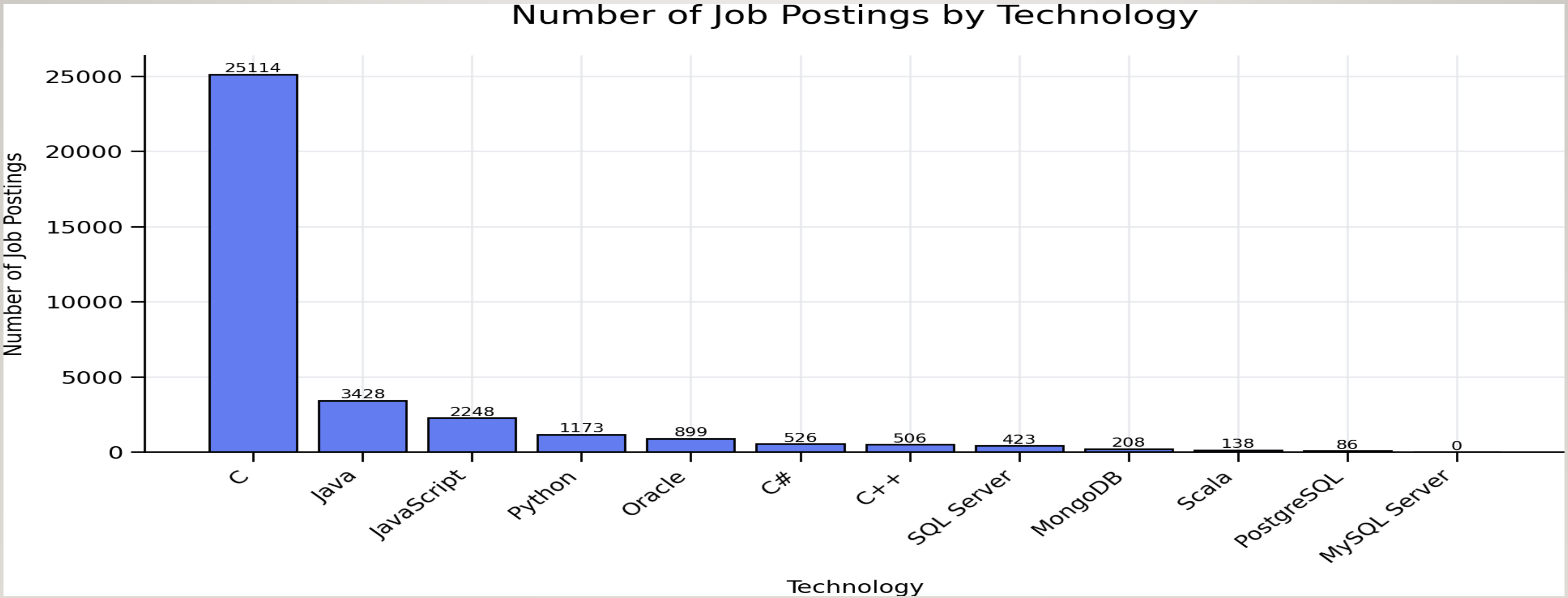
- **Dashboard Development**
- Created three dashboard tabs:
- Current Technology Usage
- Future Technology Trend
- Demographics
- **Visualization**
- Created bar charts

KEY VISUALS & DASHBOARD

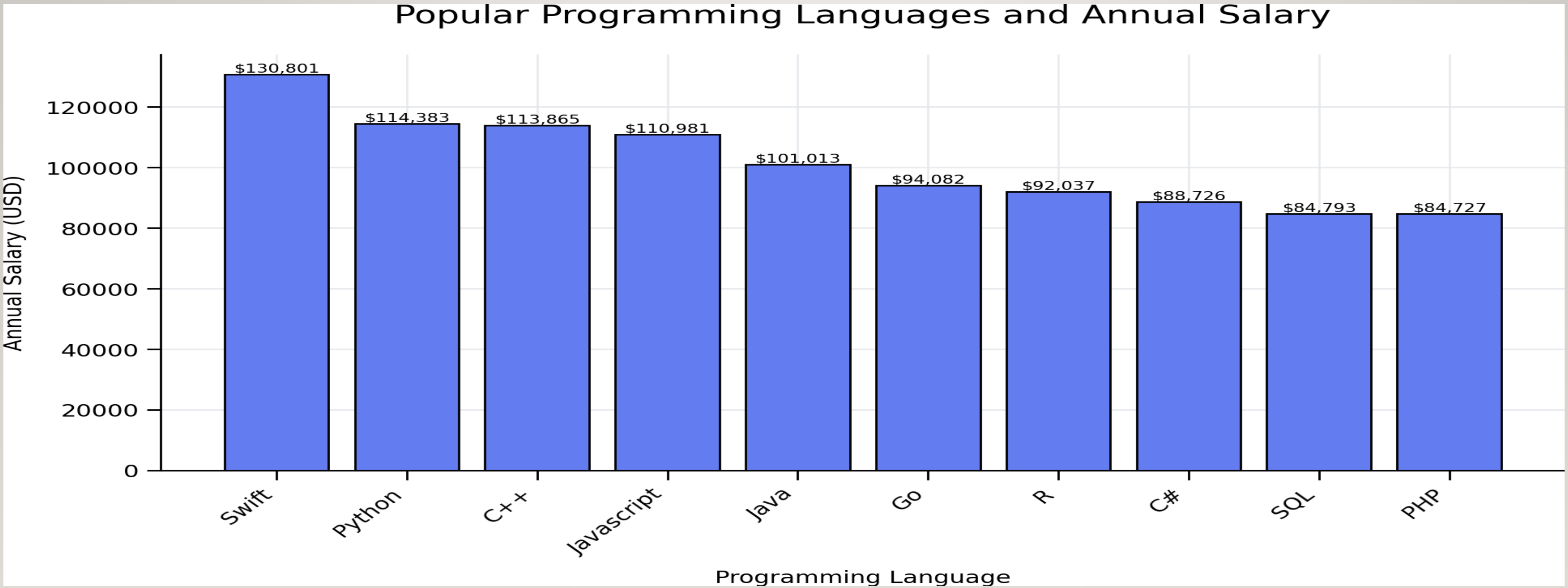
- Job Postings



NUMBER OF JOD POSTINGS BY TECHNOLOGY

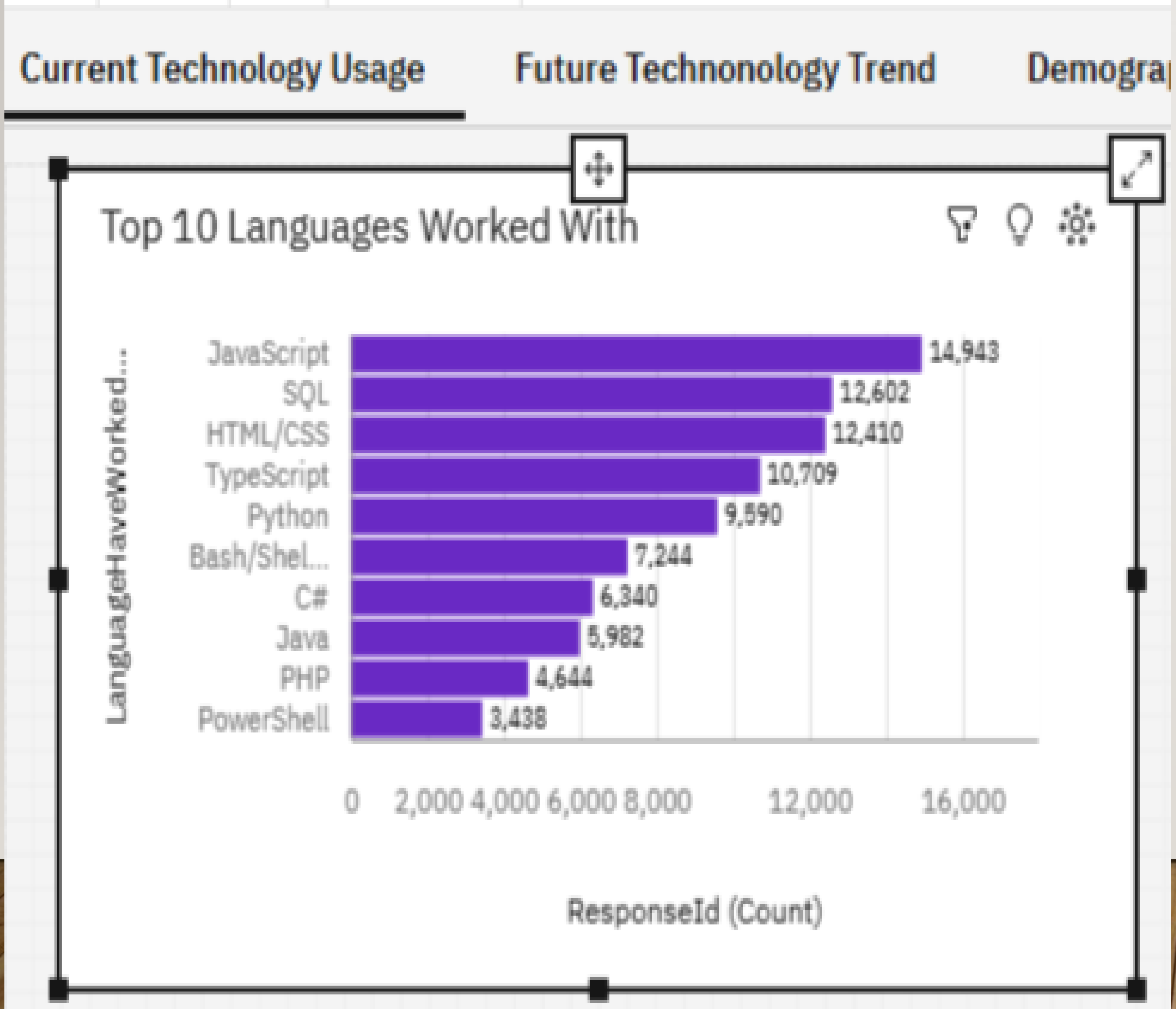


POPULAR PROGRAMMING LANGUAGE AND SALARY

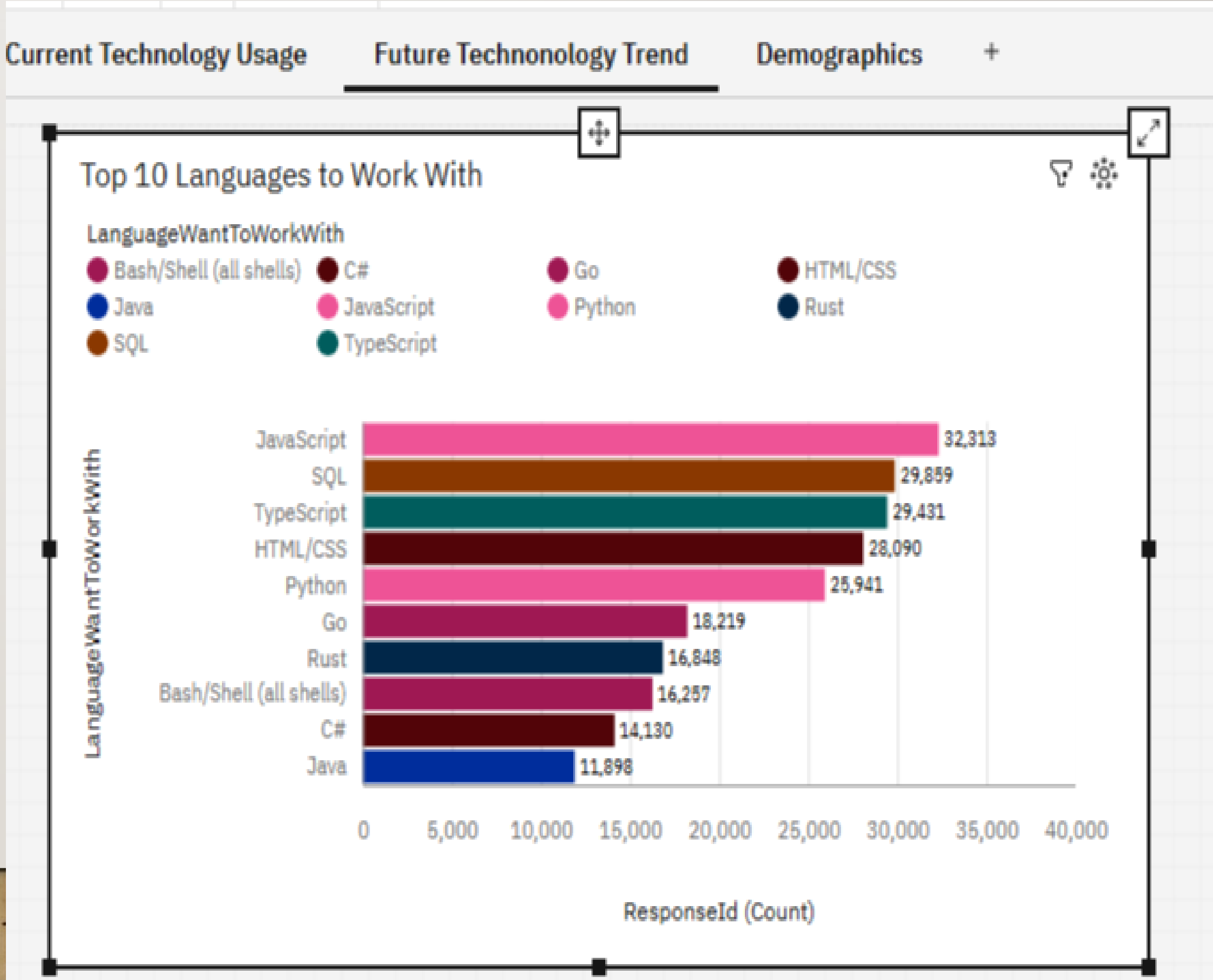


PROGRAMMING LANGUAGE TRENDS

Current Year

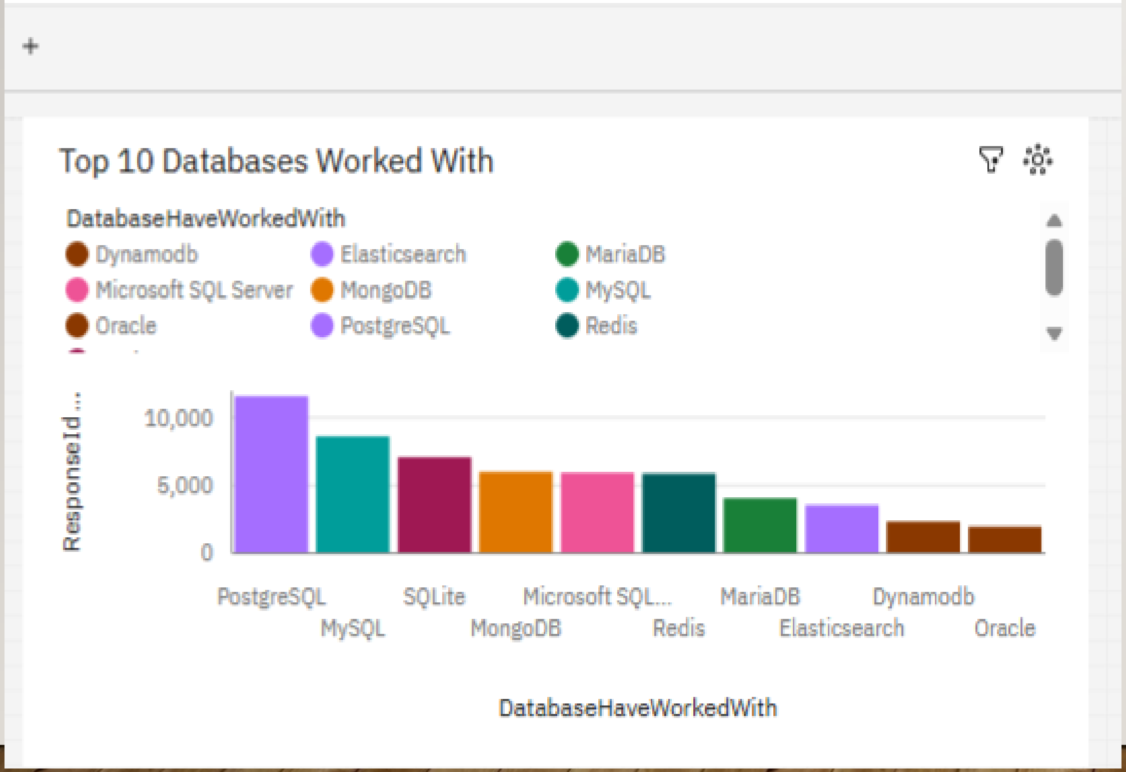


Next Year

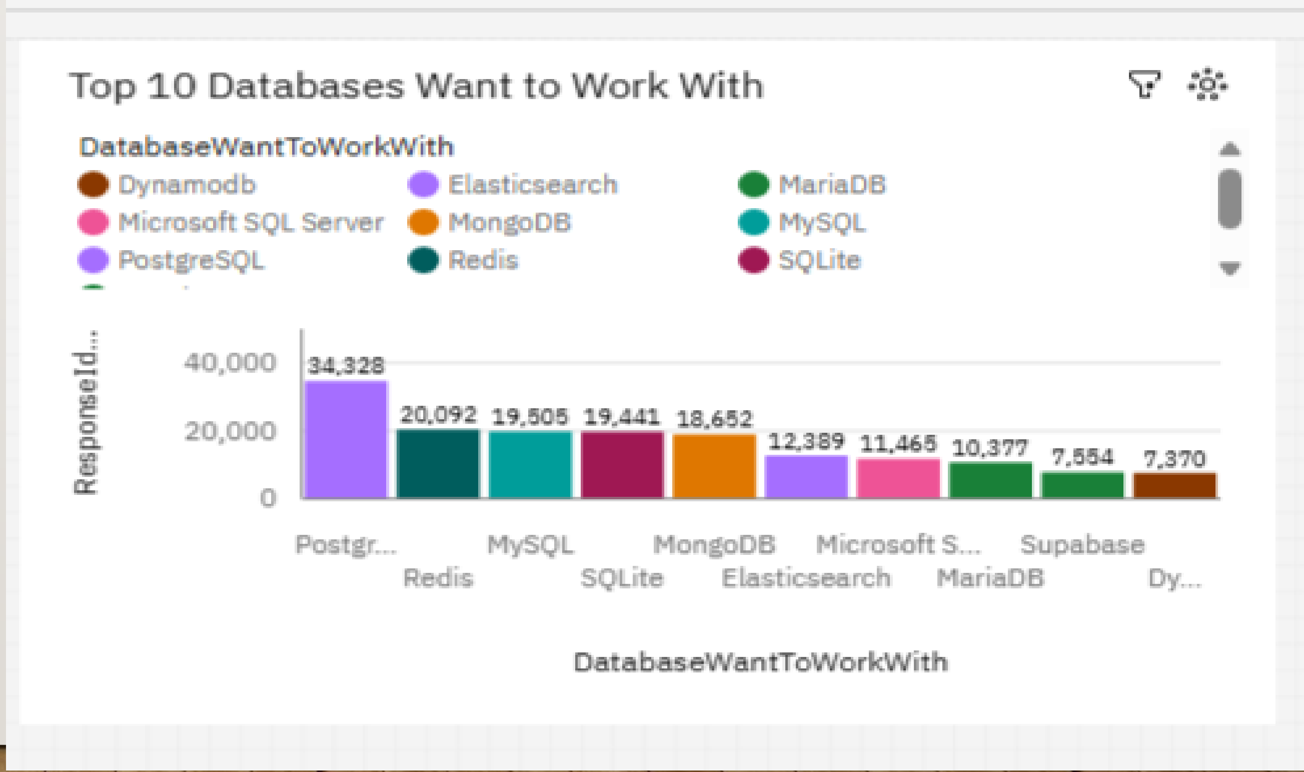


DATABASE TRENDS

Current Year



Next Year



KEY INSIGHTS

- **Key Insights**
- **Current Technology Usage**
- **JavaScript**, **SQL**, **HTML/CSS**, and **TypeScript** dominate current language usage
- **PostgreSQL** leads current database usage
- **AWS** leads current platform usage
- Mainstream cloud and web ecosystems dominate the current stack.
- **Future Technology Trend**
- **JavaScript**, **TypeScript**, and **Python** remain dominant future-interest languages
- **PostgreSQL** remains the most desired database
- **AWS** remains the strongest desired future platform
- Future demand largely reinforces current adoption while increasing interest in **TypeScript**, **Rust**, **Supabase**, and more modern web tooling.

KEY INSIGHTS

- **Demographics**
- Largest respondent age group: **25–34 (41.3%)**
- Second largest: **35–44 (27.3%)**
- Third largest: **18–24 (15.9%)**
- The strongest education profile is a **Bachelor’s degree (8,629)** followed by **Master’s degree (5,000)**.

BUSINESS IMPLICATIONS AND RECOMMENDATIONS (WHY THIS PROJECT MATTERS)

- **Programming Languages Trends**
- Organizations should continue to invest in JavaScript-based ecosystems
- Training, hiring, and curriculum planning should include stronger TypeScript capability
- Future-oriented teams may benefit from monitoring emerging demand for Rust
- **Database Trends**
- Database strategy should continue to prioritize PostgrelSQL-compatible talent and tooling
- Organizations should balance support for legacy relational databases with modern developer-first platforms
- Cloud-native and open-source database ecosystems are likely to remain influential

CHALLENGES AND HOW TO SOLVE THEM

- This project demonstrates that I can:
- Clean, messy real-world survey data,
- Transform multi-value fields into analysis-ready structures,
- Design data models for BI dashboards,
- Build interactive visual analytics in Cognos, and
- Turn raw data into decision-oriented insights.

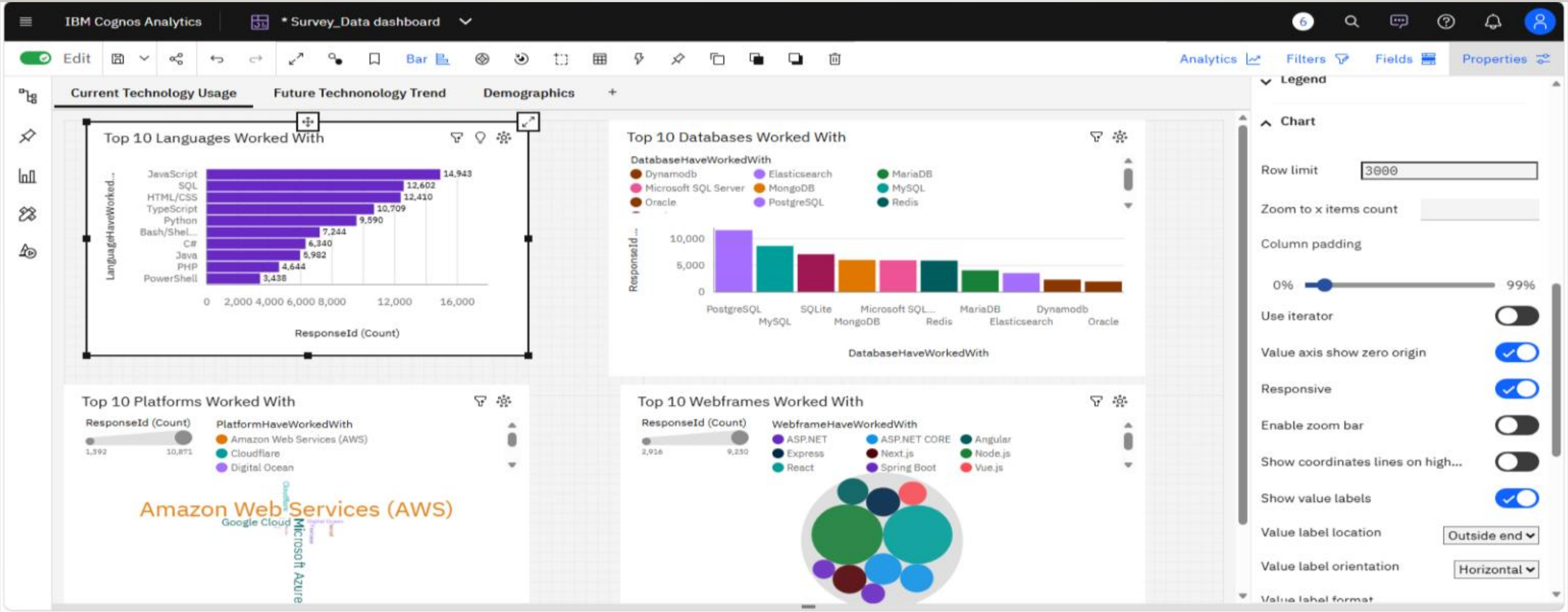
PORTFOLIO-READY "WHAT I'D IMPROVE NEXT."

- To make the project even stronger in a production setting, I would:
- automate more of the wrangling pipeline in Python,
- improve dashboard consistency and formatting,
- publish the dashboard narrative with a lighter, more executive design,
- and add a short recommendation memo for each dashboard audience.

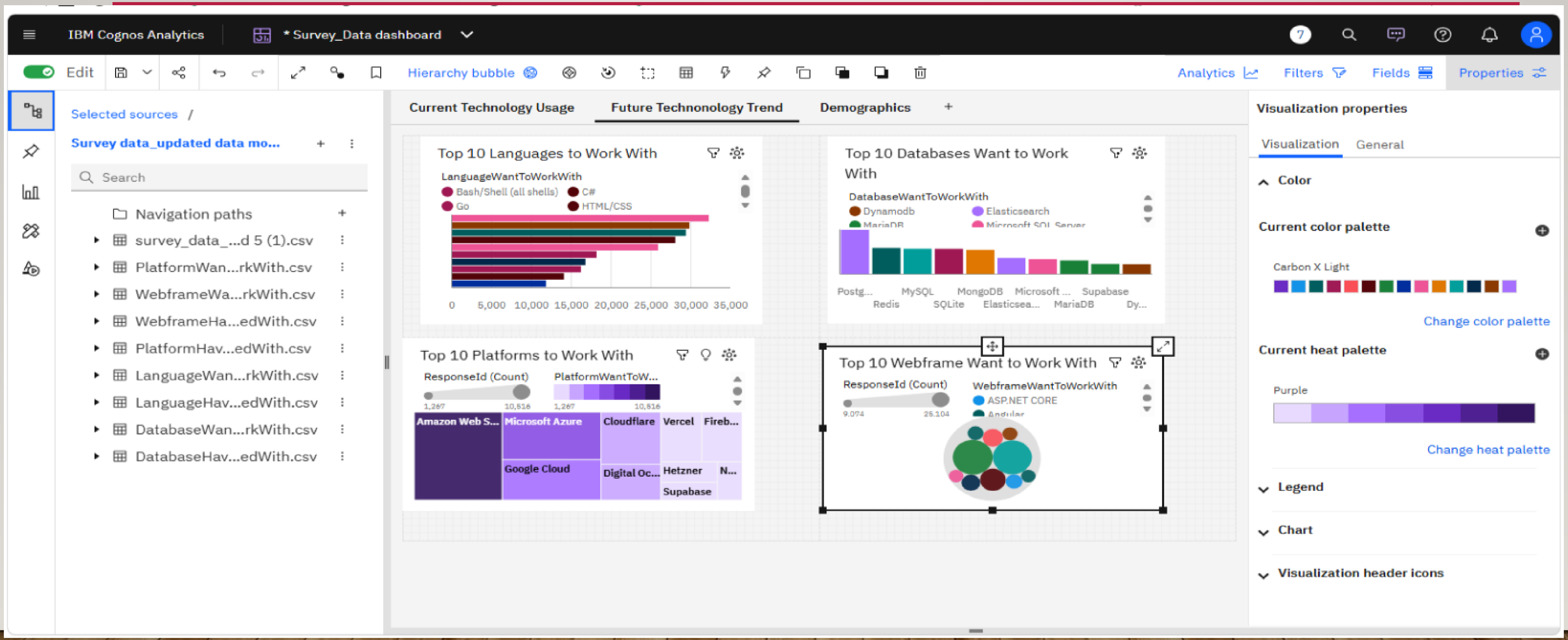
FILES AND LINKS

- notebook
- SQL script
- Dashboards
- presentation
- cleaned dataset if shareable

DASHBOARD TAB1: CURRENT TECHNOLOGY TRENDS



DASHBOARD TAB2: FUTURE TECHNOLOGY TRENDS



DASHBOARD TAB3: DEMOGRAPHICS